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Multilocation Newsvendor Problem: Centralization and Inventory Pooling

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Abstract. We study a multilocation newsvendor model with a retailer owning multiple retail stores, each of which is operated by a manager who decides the order quantity for filling random customer demand of a product. Store managers and the retailer are all risk averse, but managers are more risk averse than the retailer. We adopt conditional value-at-risk (CVaR) as the performance measure and consider two alternative strategies to improve the system's performance. First, the retailer centralizes the ordering decisions. Second, managers still decide the order quantity for their own store, whereas their inventories are pooled together. We analyze and compare the optimal order quantities and the resultant CVaR values of the systems and study their comparative statistics. For centralization, we find that each store has a higher inventory level in the centralized system than in the decentralized system, and centralization positively benefits the retailer as long as some store managers are strictly more risk averse than the retailer. When there is inventory pooling, the ordering decisions in the decentralized system depend on how the additional profit from pooling is allocated among the stores. We consider a weighted proportional allocation rule and characterize the Nash equilibrium of the resultant ordering game among the store managers. Our key finding is that as long as the store managers are sufficiently more risk averse than the retailer or the demands are very heavy tailed, inventory pooling is less beneficial than centralization. We further derive a lower bound on the value of centralization and two upper bounds on the value of inventory pooling. Finally, our analytical results are illustrated using a data set from an online retailer in China, and various comparative statics are further examined via extensive numerical experiments.

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Keywords: risk-averse newsvendor • conditional value-at-risk (CVaR) • multilocation • centralization • inventory pooling • heavy-tailed demand

1. Introduction

Consider a retailing business with multiple selling locations, for example, a traditional retailer with stores at different locations, or an e-commerce company with fulfillment centers in different cities. Currently, each store manager is responsible for deciding how much to order for a product to fill its random customer demand. The ordering behavior of the store managers exhibits risk aversion. Evidently, like other decision-makers, not all inventory managers are risk neutral; many are more concerned with the downside risk rather than the expected profit. For instance, the experimental study by Schweitzer and Cachon (2000) suggests that for the so-called "high-profit products," newsvendors exhibit risk-averse behavior. The owner of the business (the retailer) aims to improve the efficiency of this system and considers the following two strategies: first, centralizing the inventory ordering decisions, that is, a central planner will decide how much to order for each store, and second, pooling the orders/inventories of the stores together either at a physical distribution center or virtually via an information system, which will allow a store to use the other stores' excess inventory if needed.

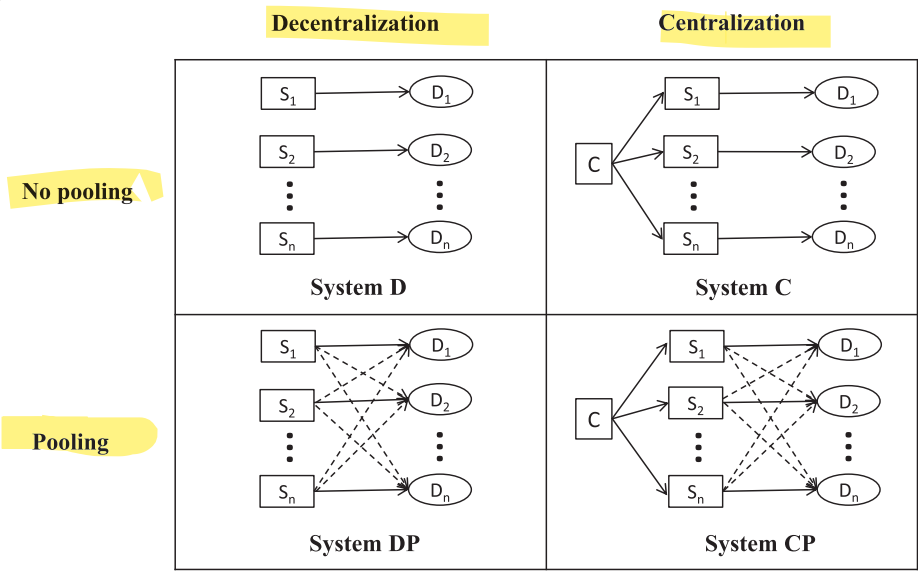
It is fairly clear that centralized ordering would benefit the retailer when managers are risk averse and that inventory pooling would also benefit the retailer by reducing the risk associated with demand uncertainty. The prior studies on multilocation/multiproduct risk-averse/risk-neutral newsvendor problems do not distinguish between centralization and inventory pooling. However, for the retailer, centralizing the inventory ordering decision and physically/virtually consolidating the stocks can be two totally different operations strategies that require different technical and managerial capabilities. Any change to convert the present system/processes to implement either one above requires a substantial investment on physical facilities, information systems, and operational capabilities. For example, to make centralized inventory ordering decision, the retailer often needs to invest in IT systems to gain visibility of the stores' inventory and develop demand forecasting capabilities; to pool inventory (physically), the retailer likely has to set up a large distribution center (DC) and to develop responsive fulfillment and logistics capabilities to fill demands in different locations. Consequently, the following managerial questions arise: First, what are the different effects of these two strategies on the inventory levels of stores? Second, how should the retailer prioritize between these two strategies if only implementing one is feasible given the available resources? Third, what are the magnitudes of values of these two strategies? The authors have recently come across several online and offline retailers in China who have been deliberating these strategic decisions, which largely motivate this research. Later in the numerical study we will use the data from one of the retailers to validate and illustrate the main results of this paper.

We consider a multilocation newsvendor model with n retail stores owned by a retailer (central planner). Each store decides the order quantity for filling its random customer demand of a product and incurs the underage or the overage costs depending on its order quantity and the realized demand. Store managers and the retailer are all risk averse, but managers are more risk averse than the retailer (some discussion of this will be provided when introducing the model details in Section 3). To model the risk aversion, we adopt conditional value-at-risk (CVaR) originally introduced in financial risk management (e.g., Rockafellar and Uryasev 2000). CVaR measures the average value of the profit falling below a certain quantile level, or VaR, which is defined as the maximum profit at a specified confidence level (see, e.g., Jorion 2006). Unlike VaR, CVaR takes into account both reward and risk and can be regarded as a special mean-risk criterion (Choi and Ruszczyński 2008). Moreover, CVaR is a coherent risk measure for quantifying risk with better computational characteristics than VaR (Artzner et al. 1999), and it is also consistent with the second-(or higher) order stochastic dominance (Ogryczak and Ruszczynski 2002). We consider two alternative strategies to improve the performance of the preceding system (system D). First, the retailer centralizes the ordering decisions (system C); that is, she decides how much to order for each store. Second, the managers still decide the order quantity for their own store whereas their inventories are pooled together so that one store facing shortage can use the other stores' excess inventory (system DP). The ideal system, where both the ordering decisions are centralized and the inventories are pooled (system CP), is also examined. The corresponding systems are illustrated in Figure 1. We analyze and compare the optimal order quantities and the resultant CVaR values of the systems and study their comparative statistics.

For centralization, we find that each store has a higher inventory level in system C than in system D. Meanwhile, centralization positively benefits the retailer as long as some store managers are strictly more risk averse than the retailer, and even when all store mangers and the retailer share the same degree of risk aversion and demands at all stores are independent, centralization could still generate benefits unless both the profit margin of the product and the number of stores are small. For inventory pooling, particularly in system DP, the ordering decisions of the store managers depend on how the additional profit generated via pooling is allocated among them, which gives rise to an ordering game among the store managers. We consider a weighted proportional allocation rule, under which each store's profit is a weighted sum of its own profit without pooling and its share of the systemwide profit. We characterize the Nash equilibrium under this allocation rule.

With the above properties, we unveil the following key insights on the values of centralization and inventory pooling: when the store managers are sufficiently more risk averse than the retailer or the demands are very heavy tailed, the central planner will prioritize centralization over inventory pooling, because the former brings more benefit than the latter. Furthermore, under some special cases, we derive bounds of the values of these two strategies. In particular, the lower bound on the value of centralization indicates that there is always value in centralization when the number of stores is large; the upper bound on the value of pooling is decreasing in all bivariate correlation coefficients of demands and if the demands are very positively correlated, then there would be no benefit from pooling. So our upper bound generalizes the main insight of Eppen (1979) to risk-averse decision-makers. We also derive an upper bound on the value of pooling for the case where the demands follow i.i.d. stable distribution,

Figure 1. Four System Structures



and the upper bound decreases as the demands become more heavy tailed.

We collect and employ data from an online retailer in China to numerically validate and illustrate our analytical results. Moreover, we use simulated data to conduct a more extensive numerical study that unveils the following insights: the benefits of centralization and inventory pooling increase as the profit margin, the number of stores, or the demand variance increases. For positively correlated demands or heavy-tailed demands, it is more likely that centralization brings more benefit than inventory pooling. Finally, we observe that centralization and inventory pooling are strategic substitutes.

The rest of the paper is organized as follows. We review the related literature in Section 2. Section 3 formulates the models in detail and provides some preliminary results. The values of centralization and inventory pooling are then investigated and compared in Section 4. We complement our analytical results with further insights from numerical study in Section 5. Finally, Section 6 concludes the paper and discusses some possible extensions. All proofs are provided in Appendix A in the electronic companion; the detailed numerical results are in Appendix B in the electronic companion; and Appendix C in the electronic companion provides the analysis for the case with exponential utility and the case with a more risk-averse retailer (than store managers).

2. Related Literature

Our paper is closely related to two streams of research: risk-averse newsvendor and inventory pooling. Both have been extensively studied.

Based on how the decision-maker's risk preference is measured, the papers on risk-averse newsvendor can be classified into three categories: utility theory (e.g., Eeckhoudt et al. 1995, Agrawal and Seshadri 2000, Chen et al. 2007, Van Mieghem 2007, Choi and Ruszczyński 2011), mean-variance (e.g., Chen and Federgruen 2000, Van Mieghem 2007, Wu et al. 2009), and VaR/CVaR (e.g., Tomlin and Wang 2005, Jammernegg and Kischka 2007, Chen et al. 2009). As our paper adopts CVaR, we focus on reviewing the papers using CVaR and discuss their differences with our paper. Jammernegg and Kischka (2007) study the risk-averse and risk-seeking newsvendor problems and conclude that a risk-averse (resp., risk-seeking) newsvendor orders less (resp., more) than a risk-neutral newsvendor. Chen et al. (2009) consider price-dependent demand and derive the optimal ordering and pricing decisions for both the additive and multiplicative demand models. For multiproduct risk-averse newsvendor models, Tomlin and Wang (2005) study how profit margin, demand correlation, resource reliability, as well as the firm's risk attitude, affect its preference of resource flexibility and supply diversification. Hanasusanto et al. (2015) develop algorithms to solve the distributionally robust optimal order quantities. The closest work to ours in this direction is Choi et al. (2011), who study a multiproduct risk-averse newsvendor under the law-invariant coherent risk measures. They show that for identical products with independent demands, increased risk aversion leads to smaller orders. Moreover, the closed-form approximations for the optimal order quantities are provided using the risk-neutral solution, which is proved to be asymptotically optimal. Note that the previous papers either consider only

a single product/location newsvendor or do not study the benefits of pooling/centralization, which is the focus of the present paper. We refer the readers to Jammernegg and Kischka (2012) for an overview of risk-averse newsvendor with VaR and CVaR.

For inventory pooling, Eppen (1979) considers a multiproduct newsvendor problem with normally distributed demands. He shows that if the inventories of the products are pooled together and their demands are independent, then the benefit from inventory pooling is of order $\sqrt{n}$, where n is the number of products. However, the benefit decreases if the demands are positively correlated, and if the demands are perfectly positively correlated, the benefit disappears. Federgruen and Zipkin (1984) consider a two-echelon distribution inventory system with positive order lead time, in which a central depot serves multiple newsvendors whereas the depot itself does not hold inventory. They provide an approximation of the optimal order quantities by the single-location inventory system and investigate the benefit of pooling for different classes of distributions. Gerchak and Mossman (1992) study cardinal effects of the extent of demand randomness on optimal inventory levels and the associated expected costs by using a mean-preserving transformation. They provide an example suggesting that in some situations risk pooling might neither reduce inventories nor move them closer to the mean or median demand. Gerchak and He (2003) study the relation between the benefit of inventory pooling and the demand variability by the mean-preserving transformation. Benjaafar et al. (2005) study the benefit from inventory pooling in production-inventory systems where the lead times are endogenously generated by a finite capacity production system. Corbett and Rajaram (2006) extend Eppen's (1979) result to more general demand distributions and prove that the benefit of inventory pooling is decreasing in the degree of positive dependence between demands at different locations. Berman et al. (2011) investigate the sensitivity of the cost reduction from inventory pooling to the demand variability and show that when the coefficient of variation of demand is low, the absolute benefit of risk pooling increases with variability, whereas, when the coefficient of variation of demand is high, both absolute and relative benefits of risk pooling decrease to zero as the demand variability increases. Aydın et al. (2012) apply the copula framework to numerically study the effects of marginal demand distributions and dependence structure on the benefit of inventory pooling. Mak and Shen (2014) investigate the benefit of inventory pooling on the degrees of positive dependence between demand and supply yield loss factors. Recently, Bimpikis and Markakis (2015) study inventory pooling under heavy-tailed demand. By considering stable distribution for demand, they show that the benefit from pooling the inventories of n products under heavy-tailed demand uncertainty is significantly lower than $\sqrt{n}$. All these works study the benefit of inventory pooling for a risk-neutral decision-maker.

Finally, we remark that in contrast to the inventory pooling literature that suggests stocking inventory at a central facility to fulfill the aggregate demands, another stream of research considers risk diversification that encourages to decentralize inventory at different locations to reduce the potential risk of disruption (see, e.g., Mak and Shen 2012, Schmitt et al. 2015). In the centralized system (without inventory pooling) we study, the inventory is also distributed at different store locations, and each store is generating its own random profit. Centralization in our model provides a way of managing a portfolio of random profits and hence results in the benefit of risk diversification even without any disruption risk. To certain extent, centralization and inventory pooling in our model also could be viewed as two alternative risk-mitigating strategies.

3. Formulation and Preliminary Results

Consider a multilocation risk-averse newsvendor problem in which there are a finite set of retail stores $\mathcal{N} = \{1, \ldots, n\}$. At the beginning of the selling season, the order quantity y_i for each retail store $i \in \mathcal{N}$ is decided for the season. The random demand at store i during the season is denoted as D_i. Let $\mathbf{y} := (y_1, \ldots, y_n)$ and $\boldsymbol{D} := (D_1, \ldots, D_n)$. We further assume that the joint probability density function for $\boldsymbol{D}$: $f(x_1, \ldots, x_n)$ is well defined (i.e., the joint distribution $F(x_1, \ldots, x_n)$ is absolutely continuous), and we denote the marginal distribution function of D_i as $F_i(\cdot)$ and its inverse function as $F_i^{-1}(\cdot)$. The selling price p and the unit ordering cost c are exogenously given and are assumed to be homogeneous across all stores. Moreover, without loss of generality, the salvage value of the leftover inventory is assumed to be zero.

We assume that the risk measures of all agents, the retailer, and the store managers are given by the conditional value-at-risk (CVaR). That is, for an agent obtaining (random) profit π, the corresponding CVaR is given by (e.g., Rockafellar and Uryasev 2000):

$$\mathrm{CVaR}_\eta(\pi) := \max_{v \in \mathbb{R}} \left\{ v - \frac{1}{\eta} \mathbb{E}[(v - \pi)^+] \right\},$$

where $\mathbb{E}$ is the expectation operator, and $\eta \in (0, 1]$ measures the degree of risk aversion of an agent: the lower η is, the more risk averse the agent (Jorion 2006, Chen et al. 2009). In particular, the agent becomes risk neutral when $\eta = 1$, and CVaR in this case reduces to the expected profit. CVaR can be interpreted as the

expected profit that falls below a certain quantile, VaR (Rockafellar and Uryasev 2000), and also can be considered to be a special case of the mean-risk criterion (Choi and Ruszczyński 2008). We denote the degree of risk aversion of store i's manager by $\hat{\eta}_i$ and that of the retailer by η. Because the retailer often can better diversify her risks associated to her relationship with the individual stores as she is engaged in many different activities (other than selling the particular product through these stores) with diverse risks, we assume that the retailer is less risk averse than the store managers, that is, $\hat{\eta}_i \le \eta$ for $i \in \mathcal{N}$. Note that in the moral hazard models in the contract theory literature, the principal is often assumed to be less risk averse than the agent (see, e.g., McMillan 1996, Salanié 2005). In particular, Salanié (2005, p. 121) states that "it is often assumed that in these models the Principal is risk neutral; for instance, the Principal faces many independent risks and thus can diversify the risks associated to his relationship with the Agent. In contrast, the Agent normally exhibits risk aversion (it is more difficult for him to diversify his risks)." Some discussion on the case when risk preferences are measured by exponential utility and the case when the retailer is more risk averse than the store managers is provided in Section 6, where we find that the main insight of this paper on comparing the values of centralization and inventory pooling continues to hold. For ease of exposition, we use $\rho(\cdot)$ and $\rho_i(\cdot)$ to represent the risk measures of the retailer and the store managers, respectively, that is, $\rho(\cdot) := \mathrm{CVaR}_{\eta}(\cdot)$ and $\rho_i(\cdot) := \mathrm{CVaR}_{\hat{\eta}_i}(\cdot)$.

The present system is decentralized where each store manager is responsible for placing orders for his own store. We consider two alternative strategies: *centralization* and *inventory pooling*. By centralization, we mean that the order quantity for each retail store $y_i, i \in \mathcal{N}$, is decided centrally by the central planner who is concerned about the overall performance of the system. In contrast, in a decentralized system, the order quantity y_i is determined by the store i's manager who is only concerned about his own performance. Inventory pooling, on the other hand, refers to the ability of fulfilling one store's demand with both its own stock and other stores' leftover stocks. Such pooling can be achieved by either holding inventory in a common warehouse (physical pooling) or allowing transshipment (when shipping costs are negligible) among the stores after the realization of demands (virtual pooling). In either case, it is natural to assume that the inventory at store i is used first to satisfy its own demand D_i, and only the leftover (if any) is used to satisfy shortages at other stores. The combination of centralization and inventory pooling strategies results in four systems: decentralized system without pooling (system D), centralized system without pooling (system C), decentralized system with pooling (system DP), and centralized system with pooling (CP). The four systems are illustrated in Figure 1.

Next, we formulate the optimization problems for the four systems and present some preliminary results of the systems.

3.1. System D

When inventory pooling is not available, store i's demand can be only met by its own stock, and its (random) profit is

$$\pi(y_i, D_i) = p(y_i \wedge D_i) - cy_i,$$

where $y_i \wedge D_i := \min\{y_i, D_i\}$, and we make the dependence on the random exogenous demand D_i explicit. Then, from the central planner's perspective, the total profit of n retail stores is given by

$$\sum_{i=1}^{n} \pi(y_i, D_i).$$

In particular, for system D, each manager of store $i \in \mathcal{N}$, decides his own order quantity y_i to maximize $\rho_i[\pi(y_i, D_i)]$. That is, the order quantity for store i is determined by

$$y_i^{\mathrm{d}} = \arg\max_{y_i} \rho_i[\pi(y_i, D_i)]. \tag{1}$$

It is well known that the optimal solution y_i^{d} in (1) admits the following closed-form solution (see, e.g., Gotoh and Takano 2007):

$$y_i^{\mathrm{d}} = F_i^{-1}(\hat{\eta}_i r), \tag{2}$$

where $r := (p - c)/p$ is the critical fractile. We also refer to r as profit margin in the following discussion. Hence, when the profit margin increases, the decentralized order quantity y_i^{d} increases, and when the degree of risk aversion of the managers increases (i.e., $\hat{\eta}_i$ decreases), the decentralized order quantity y_i^{d} decreases.

Correspondingly, the central planner obtains the value

$$\Pi^{\mathrm{d}} = \rho\left(\sum_{i=1}^{n} \pi(y_i^{\mathrm{d}}, D_i)\right). \tag{3}$$

3.2. System C

In system C, ordering decisions at all retail stores are centrally controlled by the central planner while each store meets its own demand. The optimal value of the central planner is given by

$$\Pi^{\mathrm{c}} = \max_{\mathbf{y}} \rho\left(\sum_{i=1}^{n} \pi(y_i, D_i)\right) \tag{4}$$

and the order quantities at all the stores, denoted as $\mathbf{y}^c = (y_1^c, \ldots, y_n^c)$ are the optimal solution to Problem (4). By the definition of CVaR, we can rewrite (4) as

$$\Pi^c = \max_{\mathbf{y},v}\left\{v - \frac{1}{\eta}\mathbb{E}\left[\left(v - \sum_{i=1}^{n}\pi(\mathbf{y}_i, D_i)\right)^+\right]\right\}. \tag{5}$$

Note that in the special case when all agents (the store managers as well as the retailer) are risk neutral, one can easily show that system C coincides with system D due to the linearity of the expectation. However, Problem (5) in general does not admit any closed-form solution. In fact, it is pointed out by Hanasusanto et al. (2016) that evaluating the expectation $\mathbb{E}[(v - \sum_{i=1}^{n}\pi(\mathbf{y}_i, D_i))^+]$ exactly for fixed v and $\mathbf{y}$ when $D_i, i \in \mathcal{N}$ are i.i.d. uniformly distributed is a #P-hard problem (see corollary 1 in Hanasusanto et al. 2016). Nevertheless, we manage to obtain several properties regarding the solution $\mathbf{y}^c$ as well as Π^c. It is also worth noting that Choi et al. (2011) provide an effective sample-based optimization procedure to compute the (near)-optimal order quantities and the corresponding CVaR for system C.

Proposition 1.

a. *If $\hat{\eta}_i \le \eta$ for $i \in \mathcal{N}$, then $y_i^d \le y_i^c$ and $\Pi^d \le \Pi^c$; moreover, if $\hat{\eta}_i < \eta$ for some $i \in \mathcal{N}$, then $y_i^d < y_i^c$ and $\Pi^d < \Pi^c$.*

b. *If $\hat{\eta}_i = \eta$ for all $i \in \mathcal{N}$, demands at all stores are independent, and the following inequality holds*

$$\eta + (1 - \eta r)^n \ge 1, \tag{6}$$

then $y_i^d = y_i^c = F_i^{-1}(\eta r)$ for any i, and $\Pi^d = \Pi^c$.

Proposition 1, part (a), shows that if the store managers are more risk averse than the retailer, each store has a higher inventory level in system C than in system D, and the retailer has a higher value in system C than in system D. In particular, when at least one store manager is strictly more risk averse than the retailer, centralization will always bring positive benefit to the retailer. The intuition behind the inequality $y_i^d \le y_i^c$ is clear. It is well known that a single risk-averse newsvendor tends to order less as the degree of risk aversion increases in order to protect against the downside risk. In our model, the retailer faces less downside risk than individual stores because of the benefit from pooling of random profits, even though there is no inventory pooling in this system, and she is less risk averse. As a result, she can afford to order more. Here, we point out that part (a) in Proposition 1 provides a lower bound on y_i^c, while in Choi et al. (2011) an upper bound is derived as the risk-neutral solution in the special case of i.i.d. demands, that is, $y_i^c \le F_i^{-1}(r)$. While there is no closed-form solution for y_i^c in general, with both the lower and upper bounds and the concavity, the computation for y_i^c can be greatly simplified.

Proposition 1, part (b), shows that when all the agents have an aligned attitude toward risk and Condition (6) holds, system C coincides with system D. Note that Condition (6) holds if both the number of stores n and the profit margin r are small. Thus, in this case, letting the stores operate on their own will not hurt the retailer. We remark that this result provides a complement to the main result in Choi et al. (2011). Specifically, Choi et al. (2011) propose a closed-form approximation to the solution $\mathbf{y}^c$ based on the risk-neutral solution, that is, $F_i^{-1}(r)$. The observation in Choi et al. (2011) is that the risk-neutral solution is asymptotically optimal when the number of stores $n \to \infty$. However, their approximation does not work well when n is small (see Choi et al. 2011, p. 359). Proposition 1, part (b), claims, however, that when n is small and the product has a low margin, a decentralized solution $\mathbf{y}^d$ would be optimal and system C coincides with system D.

3.3. System CP

When the retailer implements inventory pooling, from her perspective, the total profit of the system can be written as

$$\pi\left(\sum_{i=1}^{n} y_i, \sum_{i=1}^{n} D_i\right) = p\left(\sum_{i=1}^{n} y_i \wedge \sum_{i=1}^{n} D_i\right) - c\sum_{i=1}^{n} y_i.$$

When it is the central planner who decides the order quantities (system CP), the store managers do not play any role. The central planner solves the following problem:

$$\Pi^{cp} = \max_{\mathbf{y}} \rho\left(\pi\left(\sum_{i=1}^{n} y_i, \sum_{i=1}^{n} D_i\right)\right) \tag{7}$$

$$= \max_{y} \rho\left(\pi\left(y, \sum_{i=1}^{n} D_i\right)\right), \tag{8}$$

where the second equality is because only the total order quantity matters in system CP due to inventory pooling. We denote Y^{cp} as the optimal solution to (8). Letting $D_N := \sum_{i=1}^{n} D_i$, the total demand at all stores, and $F_N(\cdot)$ be the distribution of D_N, we then have $Y^{cp} = F_N^{-1}(\eta r)$.

3.4. System DP

In system DP, store managers place their own orders, whereas they will "cooperate" via inventory pooling, namely, stores facing shortage can use the other stores' excess inventory. In this case, although the store managers can unilaterally decide their own order quantities, to implement inventory pooling, they need to somehow reach an agreement with each other,

that is, determine how to allocate the residual profits from pooling. Therefore, prior to realization of demand (ex ante), the store managers first agree on an allocation rule (could be set out by the retailer in our problem), which will be used to allocate the residual profits from pooling after demand realization (ex post), and each store manager places order by considering the profit from fulfilling his own demand and potential profit from pooling. This will give rise to an ordering game among the stores. We follow and extend the work of Anupindi et al. (2001) in building the equilibrium model for system DP.

Let x_{ij} be the amount of inventory at store i that is used to satisfy the demand at store j. It then follows that for a given demand realization $\mathbf{D}$, the additional profit from inventory pooling is

$$\begin{aligned} R(\mathbf{y},\mathbf{D}) = \max_{x_{ij}\geq 0} \quad & \sum_{i\in\mathcal{N}}\sum_{j\in\mathcal{N}} p x_{ij}, \\ \text{s.t.} \quad & \sum_{i\in\mathcal{N}} x_{ij} \leq E_j, \quad \forall j\in\mathcal{N}, \\ & \sum_{j\in\mathcal{N}} x_{ij} \leq H_i, \quad \forall i\in\mathcal{N}, \end{aligned} \tag{9}$$

where $H_i = (y_i - D_i)^+$, the excess inventory after satisfying store i's own demand, and $E_i = (D_i - y_i)^+$, the excess demand at store i. Problem (9) can be solved in closed form as

$$\begin{aligned} R(\mathbf{y},\mathbf{D}) &= p\left(\sum_{i\in\mathcal{N}} H_i \wedge \sum_{i\in\mathcal{N}} E_i\right) \\ &= p\left(\sum_{i\in\mathcal{N}} (y_i - D_i)^+ \wedge \sum_{i\in\mathcal{N}} (D_i - y_i)^+\right). \end{aligned}$$

Note that $\pi(\sum_{i\in\mathcal{N}} y_i, \sum_{i\in\mathcal{N}} D_i) = \sum_{i\in\mathcal{N}} \pi(y_i, D_i) + R(\mathbf{y},\mathbf{D})$, and we will refer to $R(\mathbf{y},\mathbf{D})$ as the *residual profit*.

Let $l_i(\mathbf{y},\mathbf{D})$ be the residual profit allocated to store i. We say an allocation rule $(l_i(\mathbf{y},\mathbf{D}))_{i\in\mathcal{N}} := (l_1(\mathbf{y},\mathbf{D}),\ldots,l_n(\mathbf{y},\mathbf{D}))$ (which is a function that maps $(\mathbf{y},\mathbf{D})$ to a point in $\mathbb{R}^n$) is feasible if

$$(l_i(\mathbf{y},\mathbf{D}))_{i\in\mathcal{N}} \in \left\{(l_1,\ldots,l_n)\in\mathbb{R}^n \,\middle|\, \sum_{i\in\mathcal{N}} l_i \leq R(\mathbf{y},\mathbf{D})\right\}$$

for any $(\mathbf{y},\mathbf{D})$. The set of all feasible allocations is denoted by $\mathcal{L}$. A feasible allocation rule makes no "false promise" under any circumstance, so the total profit allocated will never exceed what is generated in the system.

Given a feasible allocation, store i decides its ordering quantity according to

$$\max_{y_i} \rho_i(\pi(y_i, D_i) + l_i(\mathbf{y},\mathbf{D})).$$

This induces a noncooperative game, and we denote its Nash equilibrium in pure strategy (if exists) by $\mathbf{y}^{\mathrm{NE}} = (y_1^{\mathrm{NE}},\ldots,y_n^{\mathrm{NE}})$. The resultant system value to the retailer is then

$$\Pi^{\mathrm{dp}} = \rho\left(\pi\left(\sum_{i=1}^n y_i^{\mathrm{NE}}, \sum_{i=1}^n D_i\right)\right). \tag{10}$$

We consider the following weighted proportional allocation rule that generalizes the proportional allocation rule studied by Anupindi et al. (2001)

$$l_i^{\mathrm{w}}(\mathbf{y},\mathbf{D}) = -w\pi(y_i, D_i) + w\left(\gamma_i \pi\left(\sum_{j\in\mathcal{N}} y_j, \sum_{j\in\mathcal{N}} D_j\right)\right),$$

where $0 \leq w \leq 1$, $\gamma_i \geq 0$ and $\sum_{i=1}^n \gamma_i = 1$. Under this allocation, each store effectively turns over a portion w of its own profit and obtains in return the same portion w of its share of systemwide profit $(\gamma_i \pi(\sum_{j\in\mathcal{N}} y_j, \sum_{j\in\mathcal{N}} D_j))$. In this way, the total profit obtained by store manager i under the weighted proportional allocation is

$$\begin{aligned} &\pi(y_i, D_i) + l_i^{\mathrm{w}}(\mathbf{y},\mathbf{D}) \\ &\quad = (1-w)\pi(y_i, D_i) + w\left(\gamma_i \pi\left(\sum_{j\in\mathcal{N}} y_j, \sum_{j\in\mathcal{N}} D_j\right)\right). \end{aligned}$$

That is, a store manager is eventually allocated a weighted sum of the profit from his own store and his share of systemwide profit. At one extreme when $w = 0$, the retailer collects all the residual profit and allocates none to the store managers. Each store manager behaves as if there were no inventory pooling at all. At the other extreme when $w = 1$, one can see that the profit obtained by each store is simply $\gamma_i \pi(\sum_{j\in\mathcal{N}} y_j, \sum_{j\in\mathcal{N}} D_j)$, which is a fraction of the systemwide profit. In this case, the allocation $l_i^{\mathrm{w}}(\mathbf{y},\mathbf{D})$ reduces to the proportional allocation rule considered in Anupindi et al. (2001) and is similar to the revenue sharing contract in the supply chain coordination literature (see, e.g., Cachon and Lariviere 2005). By considering a general weight $w \in [0,1]$, the weighted proportional allocation rule is then capable of modeling or approximating more practical allocation schemes that may lie in between the two extreme cases. Note that for any $0 \leq w \leq 1$, we have

$$\begin{aligned} \sum_{i\in\mathcal{N}} l_i^{\mathrm{w}}(\mathbf{y},\mathbf{D}) &= -w\sum_{i\in\mathcal{N}} \pi(y_i, D_i) + w\pi\left(\sum_{j\in\mathcal{N}} y_j, \sum_{j\in\mathcal{N}} D_j\right) \\ &= wR(\mathbf{y},\mathbf{D}) \leq R(\mathbf{y},\mathbf{D}), \end{aligned}$$

and hence the weighted proportional allocation is always feasible.[1]

The following proposition shows that under the weighted proportional allocation, a pure strategy Nash equilibrium always exists, and we characterize the Nash equilibrium.

Proposition 2. *Under the weighted proportional allocation, a Nash equilibrium in pure strategy, denoted by* $\mathbf{y}^{\mathrm{NE}} = (y_1^{\mathrm{NE}}, \ldots, y_n^{\mathrm{NE}})$, *exists, and*

a. *when* $w = 0$, *we have* $\mathbf{y}^{\mathrm{NE}} = \mathbf{y}^{\mathrm{d}} = (y_1^{\mathrm{d}}, \ldots, y_n^{\mathrm{d}})$;

b. *when* $w = 1$, *the equilibrium satisfies* $\sum_{i=1}^{n} y_i^{\mathrm{NE}} = F_N^{-1}(\hat{\eta} r)$, *where* $\hat{\eta} = \max_{i \in \mathcal{N}} \hat{\eta}_i$;

c. *when* $0 < w < 1$, *there exists an increasing function* $\bar{y}_i(\hat{\eta}_i)$ *with* $\lim_{\hat{\eta}_i \to 0} \bar{y}_i(\hat{\eta}_i) = 0$ *such that* $y_i^{\mathrm{NE}} \le \bar{y}_i(\hat{\eta}_i)$.

Although in Proposition 2, parts (a) and (b), we have a closed-form characterization of Nash equilibrium for the two extreme cases (in particular, for part (b) when $w = 1$, one equilibrium is that the least risk-averse store manager would order $F_N^{-1}(\hat{\eta} r)$ for all the stores, while the other stores order nothing), the ordering game among the stores is in general not supermodular, and so the standard techniques used to perform comparative statics for Nash equilibrium do not directly apply here (see, for instance, Topkis 2011 for a comprehensive account on supermodular games). Nevertheless, Proposition 2, part (c), shows that a store's equilibrium order quantity is bounded by a function that is decreasing toward zero as the store becomes more risk averse.

We remark that under the weighted proportional allocation, it is possible that $l_i^{\mathrm{w}}(\mathbf{y}, \mathbf{D}) < 0$ for some store i and demand realization $\mathbf{D}$ as the following example shows.

Example 1. Consider $w = 1, p = \$2, c = \$1$ and $\mathbf{y} = (3, 8), \mathbf{D} = (4, 0)$. The systemwide profit is shared equally between the two stores, that is, $\gamma_1 = \gamma_2 = 1/2$. In this case, store 1 makes a profit of \$3 on its own, whereas store 2 loses \$8. The residual profit from pooling is $R(\mathbf{y}, \mathbf{D}) = \$2$. The retailer is in total losing a profit of \$3. Hence, according to the weighted proportional allocation rule, each store is responsible for $-\$1.5$, and the corresponding allocations of the residual profit are $l_1^{\mathrm{w}}(\mathbf{y}, \mathbf{D}) = -4.5$ and $l_2^{\mathrm{w}}(\mathbf{y}, \mathbf{D}) = 6.5$. One can see that store 1 is compensating for the loss at store 2 in this particular instance.

A negative allocation would potentially deter the stores from participating in inventory pooling. One remedy to this issue is to add a strategy-independent side payment $s_i(\mathbf{D})$ with $\sum_{i \in \mathcal{N}} s_i(\mathbf{D}) = 0$ to the weighted proportional allocation such that $\tilde{l}_i^{\mathrm{w}}(\mathbf{y}, \mathbf{D}) = l_i^{\mathrm{w}}(\mathbf{y}, \mathbf{D}) + s_i(\mathbf{D}) \ge 0$. It can be shown by using a similar approach as in Anupindi et al. (2001) that such side payments always exist.

4. Value of Centralization and Pooling

In this section, we study the implications of centralization and inventory pooling on the performance of the systems. Specifically, we seek to understand for a central planner operating system D, how she should prioritize between centralizing the ordering decisions and building up a system that allows for inventory pooling. In other words, given that $\Pi^{\mathrm{d}} \le \Pi^{\mathrm{dp}}$ and $\Pi^{\mathrm{d}} \le \Pi^{\mathrm{c}}$, that is, both centralization and pooling bring benefit, when will $\Pi^{\mathrm{dp}} < \Pi^{\mathrm{c}}$ or $\Pi^{\mathrm{dp}} > \Pi^{\mathrm{c}}$? The following theorem summarizes our finding.

Theorem 1. *Under the weighted proportional allocation, there exists* $\hat{\eta}_w \le \eta$ *such that if* $\hat{\eta}_i \in (0, \hat{\eta}_w)$ *for all* $i \in \mathcal{N}$, *then* $\Pi^{\mathrm{dp}} < \Pi^{\mathrm{c}}$.

Theorem 1 shows that under the weighted proportional allocation as long as the store managers are sufficiently risk averse, it is beneficial to first centralize the ordering decisions rather than pooling the inventory. Intuitively, when the managers are very risk averse, they tend to order too few, which results in a low system inventory level, and so the benefit of inventory pooling is minimal.

The benefit of inventory pooling, as noted by Bimpikis and Markakis (2015) in the risk-neutral case, is greatly affected by the tail behavior of the demand distribution. Particularly the benefit of pooling under heavy-tailed demand uncertainty can be significantly lower than that for normally distributed demand (light tailed). This implies that the tail behavior of demand distribution may affect the priority between centralization and pooling. Given recent empirical evidences on strong heavy-tailed phenomena in many industries (see, e.g., Chevalier and Goolsbee 2003, Gaffeo et al. 2008, Bimpikis and Markakis 2015), it is important to understand the impact of the tail behavior of the demand distribution on the value of centralization and pooling in our risk-averse setting. To study this issue, we next assume that the demands are i.i.d. according to a *stable distribution*. A stable distribution, denoted as $S_\alpha(\sigma, \beta, \mu)$, is characterized by four parameters: the stability parameter $\alpha \in (0, 2]$, the skewness parameter $\beta \in [-1, 1]$, the scale parameter $\sigma > 0$, and the location parameter μ. More specifically, α determines the shape of the tail of the distributions. The distributions have undefined variance for $\alpha < 2$, and undefined mean for $\alpha \le 1$; β determines how skewed the distribution is and to which direction; σ corresponds to the standard deviation whenever it is finite; and, finally, μ corresponds to the mean whenever it is finite. In particular, when $\alpha = 2$ and $\beta = 0$, the distribution reduces to the normal distribution, and, as α decreases, the distribution becomes more heavy tailed: when $\alpha = 1$, the distribution becomes the Cauchy distribution.

Theorem 2. *Let* $S_\alpha(\sigma, \beta, \mu)$ *be a stable distribution. Suppose demands at all the stores are i.i.d. with* $D_i \sim S_\alpha(\sigma, \beta, \mu)$, *and* $\hat{\eta}_i = \hat{\eta} < \eta$ *for all* $i \in \mathcal{N}$. *Consider a symmetric equilibrium under the weighted proportional allocation for system DP. Then, there exists* $\alpha_w \in (1, 2]$, *such that for any* $\alpha \in (1, \alpha_w]$, $\Pi^{\mathrm{dp}} \le \Pi^{\mathrm{c}}$.

Theorem 2 posits that when the demands are very heavy tailed, then it is beneficial to first centralize the ordering decisions rather than pooling the inventory. The underlying reason is that when the demands are heavy tailed, the benefit of inventory pooling is rather limited as we will show in Proposition 5, and, thus, the benefit of centralization dominates.

Finally, we remark that Alfaro and Corbett (2003) study a related question in a risk-neutral setting: will inventory pooling still generate benefits under a suboptimal inventory level? They also discuss whether one should prioritize on optimizing the inventory level in a dedicated (decentralized) system or building a pooling system under the suboptimal inventory level. We point out two key differences of our paper from theirs here. First, we consider a more general model than Alfaro and Corbett (2003) by allowing the agents involved to be risk averse and by distinguishing between the notion of "centralization" and "pooling." Second, while the system inventory level in the DP system $Y^{\mathrm{dp}} := \sum_{i=1}^{n} y_i^{\mathrm{NE}}$ can be interpreted as a suboptimal inventory level from the central planner's perspective, in our model, it is endogenously determined and depends on the agent's degree of risk aversion. In contrast, a suboptimal inventory level is given as an exogenous parameter in Alfaro and Corbett (2003).

4.1. Magnitude of the Benefits

The preceding results compare the values of centralization and inventory pooling. Next, we further study the magnitudes of the values of these two strategies. It is, in general, difficult to quantify these values analytically. In the following, we provide bounds on the values of centralization and inventory pooling for certain special cases. We first examine the value of centralization by considering independent demands.

Denote the risk-neutral optimal order quantities as $y_i^0 = F_i^{-1}(r)$. Note that Choi et al. (2011) prove that (a) $y_i^{\mathrm{c}} \le y_i^0$, $i = 1, \ldots, n$ and that (b) $\mathbf{y}^0$ is asymptotically optimal for system C as the number of stores approaches infinity. Intuitively, when the number of stores is sufficiently large, by the central limit theorem, the volatility (risk) of the total profit becomes negligible compared with the mean value of the total profit. As a result, the risk-neutral solution that maximizes the mean value of the total profit achieves the near-optimal CVaR value.

To establish the asymptotic result, similar to Choi et al. (2011), we need the following two mild assumptions, to guarantee that the central limit theorem holds.

i. There exist $y^{\min}$ and $y^{\max}$ such that

$$y^{\min} \le y_i^{\mathrm{d}} \le y_i^0 \le y^{\max}, i = 1, \ldots, n.$$

ii. There exists $\sigma_{\min} > 0$ such that

$$\min\left(\mathbb{V}ar\left[\min\left(y_i^0, D_i\right)\right], \mathbb{V}ar\left[\min\left(y_i^{\mathrm{d}}, D_i\right)\right]\right) \ge \sigma_{\min}^2, \ i = 1, \ldots, n.$$

Let $f_i(x)$ denote the marginal density function of D_i, we have the following result:

Proposition 3. *Under assumptions* (i) *and* (ii), *we have*

$$\frac{1}{n}\left(\Pi^{\mathrm{c}} - \Pi^{\mathrm{d}}\right) \ge \frac{p}{2} \min_{i=1,\ldots,n} \left\{ \min_{x \in [y_i^{\mathrm{d}}, y_i^0]} \{f_i(x)\}\left(y_i^0 - y_i^{\mathrm{d}}\right)^2 \right\} - O\left(1/\sqrt{n}\right). \tag{11}$$

This result shows that the benefit of centralization is of order n if the decentralized order quantities of the stores are strictly different from those of the risk-neutral optimal order quantities and the density function of the demand of each store i is strictly positive within the interval $[y_i^{\mathrm{d}}, y_i^0]$. The intuition behind this result is that the gap in CVaR between system D and system C is mainly because of the disparity in order quantities resulted from the misaligned risk preferences between the retailer and the store managers. Therefore, the benefit of centralization mainly depends on the gap between the optimal order quantities of the two. Meanwhile, when the number of stores is large, the optimal order quantities of system C are close to the risk-neutral optimal order quantities. Consequently, the benefit of centralization can be captured by the gap between the risk-neutral optimal order quantities and the decentralized optimal order quantities. Note that as long as $\hat{\eta}_i < 1$ for $i = 1, \ldots, n$, $y_i^{\mathrm{d}} < y_i^0$, and so (11) provides a nontrivial lower bound on the value of centralization asymptotically. In particular, when $\hat{\eta}_i = \eta < 1$, (11) implies that there is always strictly positive value in centralization in an asymptotic sense even if the risk preferences of the retailer and store managers are aligned. This complements part (b) of Proposition 1, which shows there could be no value in centralization when the number of stores is small, that is, $\Pi^{\mathrm{d}} = \Pi^{\mathrm{c}}$ if $\eta + (1 - \eta r)^n \ge 1$ (note that when n is sufficiently large, this condition will be violated). Note that here demands are assumed to be independent. When demands are correlated and identically distributed, we can obtain a similar lower bound on the value of centralization under some convergence conditions on the sequence of correlation coefficients, for example, $\lim_{n\to\infty} \sum_{1\le i<j\le n} |\rho_{ij}^k| < \infty$ for $k = 0,\mathrm{d}$ where ρ_{ij}^k is the correlation coefficient between $\min\{y_i^k, D_i\}$ and $\min\{y_j^k, D_j\}$ (e.g., Fan and Yao 2013, pp. 74–75).

Eppen (1979) and Corbett and Rajaram (2006) show that demand dependence greatly affects the value of

inventory pooling when managers are risk neutral. The next result studies the benefit of inventory pooling under the multivariate normal distribution of demands when the proportional allocation is used in system DP ($w = 1$). We aim to investigate the effect of the correlation between demands on the value of inventory pooling under the risk-averse setting.

Proposition 4. *When $\hat{\eta}_i = \hat{\eta}$ for $i = 1, \ldots, n$ and $\boldsymbol{D}$ follows a multivariate normal distribution with (μ_i, σ_i) being the mean and standard deviation of D_i and ρ_{ij} being the correlation coefficient between D_i and D_j, and the proportional allocation is used in system DP ($w = 1$), we have*

$$\Pi^{\mathrm{dp}} - \Pi^{\mathrm{d}} \le \left(\sum_{i=1}^{n} \sigma_i - \sigma_N\right) \cdot \left[c\Phi^{-1}(\hat{\eta}r) - \rho\left(p \min\left(\Phi^{-1}(\hat{\eta}r), X\right)\right)\right], \tag{12}$$

where $\sigma_N = \sqrt{\sum_{i,j\in\mathcal{N}} \sigma_i\sigma_j\rho_{ij}}$, $\Phi^{-1}(\cdot)$ is the inverse function of the standard normal distribution and X is a standard normal random variable.

Note that $\sum_{i=1}^{n} \sigma_i \ge \sigma_N$ and the equality holds if and only if $\rho_{ij} = 1$ for all $i \ne j$. Because the benefit of inventory pooling cannot be negative, the values of $c\Phi^{-1}(\hat{\eta}r) - \rho(p \min(\Phi^{-1}(\hat{\eta}r), X))$ under the parameters of interest are positive. Also note that when the store managers are all risk neutral, then (12) becomes an equality. The implication of the above proposition is similar to that of Eppen (1979). It posits that when demands follow a multivariate normal distribution, the upper bound of the benefit of inventory pooling is decreasing in all bivariate correlation coefficient ρ_{ij}. In particular, if the demands are perfectly positively correlated, that is, $\rho_{ij} = 1$, then there will be no benefit of inventory pooling. And if $\sigma_i = \sigma$ for all i and $\rho_{ij} = \varrho$ for all $i \ne j$, then we have $\sum_{i=1}^{n} \sigma_i - \sigma_N = \sigma(n - \sqrt{n + n(n-1)\varrho})$ which is decreasing in ϱ.

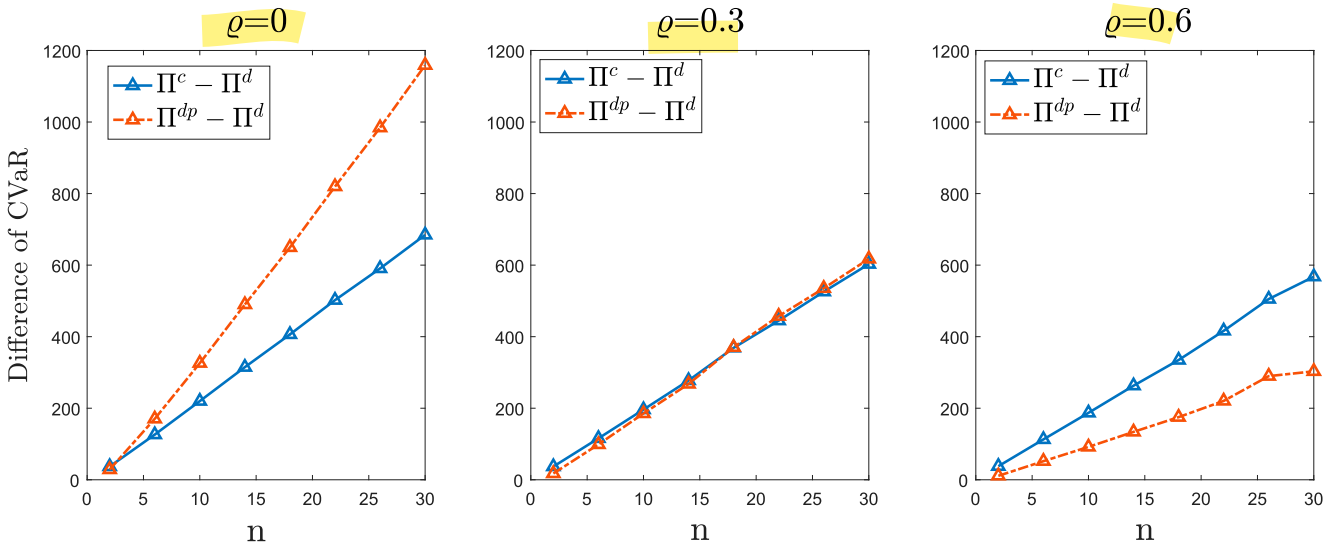
Figure 2 shows how $\Pi^{\mathrm{c}} - \Pi^{\mathrm{d}}$ and $\Pi^{\mathrm{dp}} - \Pi^{\mathrm{d}}$ depend on the number of stores n when demands follow multivariate normal distributions with $\mu_i = 10$ and $\sigma_i = 4$ for $i = 1, \ldots, n$ and $\rho_{ij} = \varrho$ for all $i \ne j$ and the proportional allocation is used. Here, we vary ϱ at 0, 0.3, and 0.6 and fix the other parameters at $\eta = 0.9$, $\hat{\eta}_i = 0.4$, $p = 20$ and $c = 4$. We can see from the figure that both the benefits of centralization and inventory pooling are almost linear in n. Meanwhile, from these examples, we can observe that the benefit of inventory pooling greatly declines as the correlation coefficient of demands increases while the benefit of centralization is not sensitive to the demand correlation. This validates the theoretical bound on the benefits of inventory pooling shown in Proposition 4. Therefore, for positively correlated demands, it is more likely that the retailer should prioritize centralization over inventory pooling.

The next result studies the magnitude of the benefit of inventory pooling under the stable distribution of demands when the proportional allocation is used in system DP.

Proposition 5. *When $\hat{\eta}_i = \hat{\eta}$ for $i = 1, \ldots, n$ and the demands are i.i.d. stable random variable $S_\alpha(\sigma, \beta, \mu)$, and the proportional allocation is used in system DP ($w = 1$),*

$$\frac{1}{n}\left(\Pi^{\mathrm{dp}} - \Pi^{\mathrm{d}}\right) \le \left(1 - n^{1/\alpha - 1}\right)\left[(p - c)\mu - \rho\left(\pi\left(y^{\mathrm{d}}, D\right)\right)\right],$$

Figure 2. (Color online) Benefits of Centralization and Inventory Pooling under Different Levels of Demand Correlation



Notes. The benefits of centralization and pooling are almost linear in n. The benefit of pooling declines as the correlation coefficient increases, whereas the benefit of centralization is not sensitive to the demand correlation.

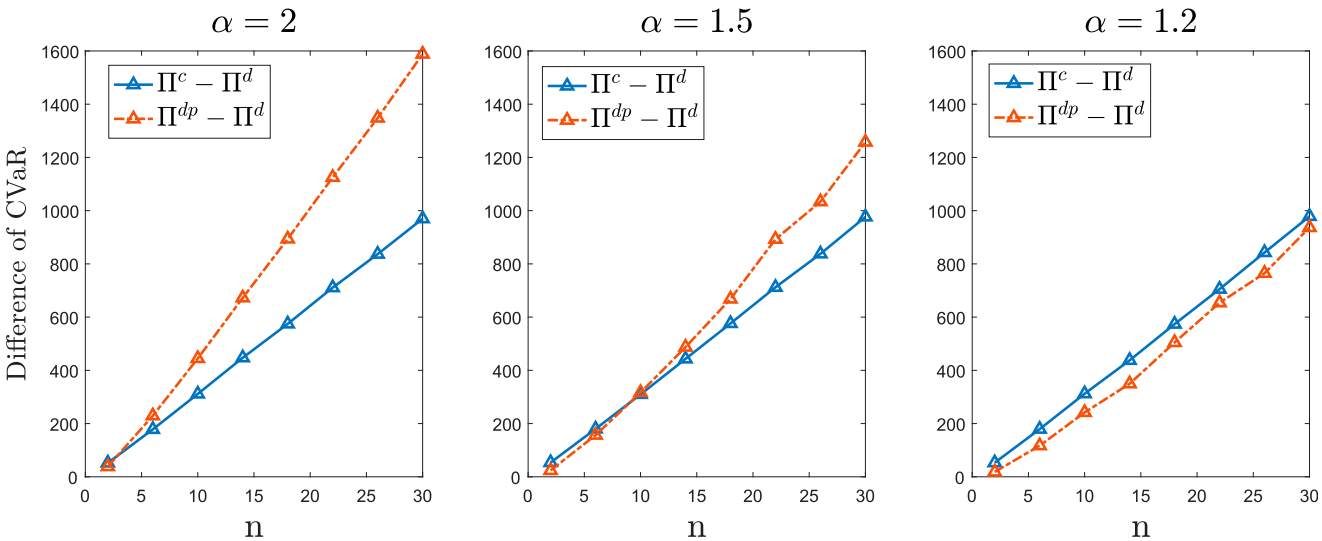
where y^{d} and D are any store's order quantity and demand, respectively.

Proposition 5 provides an upper bound on the benefit of inventory pooling, and we note that the above bound is tight when the retailer is risk neutral. It can be seen that this upper bound is of order n as long as $\alpha > 1$, and the coefficient of n is determined by the gap between the deterministic profit and the CVaR value of the newsvendor profit. As α decreases (approaches to 1), that is, demands become more heavy tailed, $(1 - n^{1/\alpha-1})$ decreases. In particular, there would be no benefit of inventory pooling when $\alpha = 1$; that is, demands follow Cauchy distribution. Meanwhile, α also affects $\rho(\pi(y^{\mathrm{d}}, D))$, and we can expect that it increases in α, and thus the coefficient $[(p-c)\mu - \rho(\pi(y_i^{\mathrm{d}}, D_i))]$ decreases in α. However, as noted by Bimpikis and Markakis (2015), the impact of α on $[(p-c)\mu - \rho(\pi(y_i^{\mathrm{d}}, D_i))]$ should not scale with n, that is, as α decreases, the upper bound on the benefit of inventory pooling decreases.

Figure 3 shows how $\Pi^{\mathrm{c}} - \Pi^{\mathrm{d}}$ and $\Pi^{\mathrm{dp}} - \Pi^{\mathrm{d}}$ depend on the number of stores n when demands follow i.i.d. stable distribution. Here, we vary α at 2, 1.5, and 1.2 and fix the other parameters at $\eta = 0.9$, $\hat{\eta}_i = 0.4$, $p = 20$, $c = 4$, $\mu = 10$, $\sigma = 4$ and $\beta = 0$. From these examples, we can observe that when the demand is light tailed (e.g., $\alpha = 2$) and the number of stores is relatively small, the benefit of centralization dominates the benefit of inventory pooling, whereas, as the number of stores increases, the benefit of inventory pooling increases faster and becomes higher than the benefit of centralization. When the demand distribution becomes more heavy tailed (i.e., α deceases from 2 to 1.2), the benefit of inventory pooling increases slower as n increases. When the demand is very heavy tailed ($\alpha = 1.2$), the benefit of centralization always dominates the benefit of inventory pooling. Therefore, for heavy-tailed demand, it is more likely that the retailer will prioritize centralization over inventory pooling, which is consistent with our result in Theorem 2.

4.1.1. Remark. We have discussed the values of centralization and inventory pooling so far without considering possible costs that may incur in practice when implementing either strategy, that is, investment in IT systems, facilities, and managerial and transportation costs. IT systems play an important role in collecting and analyzing information, for example, demand forecast, of the store network when implementing either strategy. Usually the cost of IT systems is more functional driven and is less affected by the number of stores managed by the retailer, and thus it can be viewed as sort of fixed costs. Hence, the benefits of both strategies would exceed the cost of installing the IT systems as long as the number of stores is sufficiently large. Meanwhile, inventory pooling can be achieved either virtually or physically. Pooling inventory virtually incurs additional transhipment/transportation costs, whereas physically pooling inventory means fewer but larger facilities, and each facility would cover demand in a larger geographical region and likely increase the transportation costs. Therefore, additional transportation costs would reduce the benefit of inventory pooling and potentially lower its strategic priority.

Figure 3. (Color online) Benefits of Centralization and Inventory Pooling under Different Tail Behaviors of Demands



Note. When the tail of demand distribution is heavier (α is smaller), the benefit of pooling decreases, whereas the benefit of centralization is not greatly affected.

5. Numerical Studies

In this section, we first conduct a numerical study with problem parameters calibrated by real data we collected from an online retailer in China to illustrate and validate our theoretical results. Then we provide a summary of the observations from a more comprehensive numerical study using simulated parameters. Because of space limits, the detailed numerical results are given in Appendix B in the electronic companion.

As we have discussed in Section 3.2, the stochastic optimization problem (4) is difficult to solve exactly. Here, we follow Choi et al. (2011) by using the sample-based optimization procedure to compute the optimal order quantities and the corresponding CVaR for system C. Its details are also given in Appendix B in the electronic companion. Readers are referred to Shapiro (2008) for the application of sample-based optimization procedure for more general risk-averse stochastic programs and its asymptotic properties.

5.1. Practice-Based Validation Study

We first calibrate the problem parameters using the retail data collected from an online retailer in China and then use the numerical results to illustrate and validate our theoretical results. The online retailer serves several regions in China. The demands of each region are fulfilled by the corresponding distribution center (DC) that is managed separately. We focus here on two of its DCs located in North China that are geographically close to each other. Two stock-keeping units (SKUs) are randomly selected under our study with one being a pet food and the other one being a body care product. The average price and cost of the former are 105 CNY (1 USD≈ 6.9 CNY) and 23.5 CNY, respectively, and those of the latter are 102 CNY and 45 CNY. We collect the daily demand data of these two SKUs in the DCs from August 1, 2017 to December 16, 2017, and obtain 128 data points for each (SKU, DC) pair. For each SKU, we use the inbound data of the DCs to keep track of their daily inventory status. On average, the order cycle at each of the DCs is around two weeks. The data points are removed from our study whenever one of the DCs was out of stock (in which case demand is censored). Finally, we keep 112 and 94 data pairs for SKU1 and SKU2, respectively.

Because the observed order cycles at the DCs are around two weeks, we calibrate a newsvendor model with biweekly demands. Directly aggregating the daily demand data would result in less than 10 samples for the biweekly demand. We overcome this problem by bootstrapping from the daily demand before aggregating them into biweekly demand. Specifically, for each daily demand sample, we drew randomly additional 13 daily demand samples with replacement from the empirical distribution of the daily demand. The 14 daily demand samples are aggregated to obtain one sample for the biweekly demand. With bootstrapping, in total we obtain 112 and 94 biweekly demand samples for SKU1 and SKU2, respectively (see Figure 4).

One can see from Figure 4 that the demands at the two DCs are positively correlated (the empirical coefficients of correlation (rank correlation) are 0.5161 and 0.3721 for SKU1 and SKU2, respectively). Because of the limited number of samples, directly fitting the joint demand distribution is difficult. Here, we use the copula approach (readers are referred to Aydın et al. 2012 for detailed discussions on the copula approach in the inventory pooling problems), to estimate the joint distribution of the demands at the two DCs. In particular, a two-dimensional copula $C(\cdot,\cdot):[0,1]\times[0,1]\to[0,1]$ for a random vector (X_1,X_2) is defined as a function that satisfies $C(F_1(x_1),F_2(x_2))=F(x_1,x_2)$, where $F(\cdot,\cdot)$ is the joint distribution of (X_1,X_2) and $F_i(\cdot)$ is the marginal distribution of X_i, $i=1,2$. A commonly used copula is the bivariate Gaussian copula (see Meyer 2013 for a comprehensive discussion) defined as

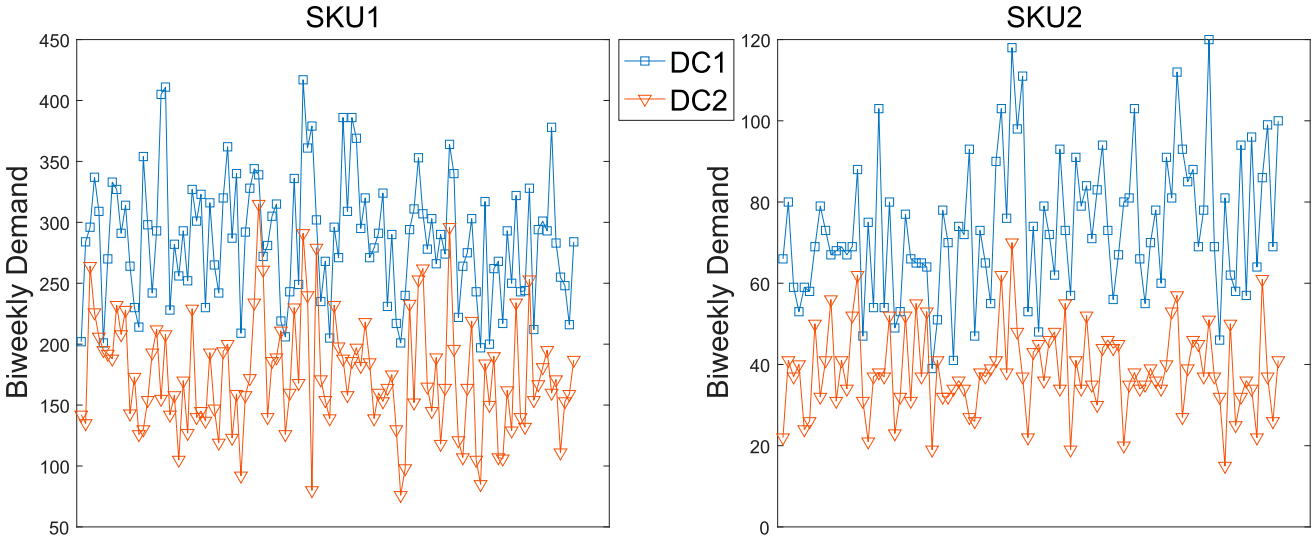
$$C(u,v;\varrho):=\Phi_2\left(\Phi^{-1}(u),\Phi^{-1}(v);\varrho\right),$$

where $\Phi_2(\cdot,\cdot)$ is the distribution function of the standard bivariate normal random variable with correlation of coefficient ϱ and $\Phi(\cdot)$ is the standard univariate normal distribution. In our study, we use the bivariate Gaussian copula with the empirical marginal distribution $\hat{F}_i(x_i)$ to estimate the joint distribution, that is, $\hat{F}(x_1,x_2)=C(\hat{F}_1(x_1),\hat{F}_2(x_2);\varrho)$. The parameter ϱ in the Gaussian copula is estimated from the data such that the coefficient of correlation of the estimated distribution matches the empirical coefficient of correlation.[2] To sample from this joint distribution, we use the following procedure: (1) generate a bivariate normal random variable (X_1,X_2) with mean 0 and covariance matrix $\Sigma=\left[\begin{smallmatrix}1 & \varrho\\ \varrho & 1\end{smallmatrix}\right]$; (2) let $U_i=\Phi(X_i)$, $i=1,2$; and (3) let $D_i=\hat{F}_i^{-1}(U_i)$.

With the demand samples and the price and cost information of the SKUs, the sample-based optimization procedure is applied to obtain the optimal order quantities and the corresponding CVaR of the four systems as the degree of risk aversion of the retailer varies (for system DP, we only focus on the results under the proportional allocation; $w=1$). The degree of risk aversion of the managers of DCs are fixed at 0.4. The results of SKU1 and SKU2 are shown in Figure 5.

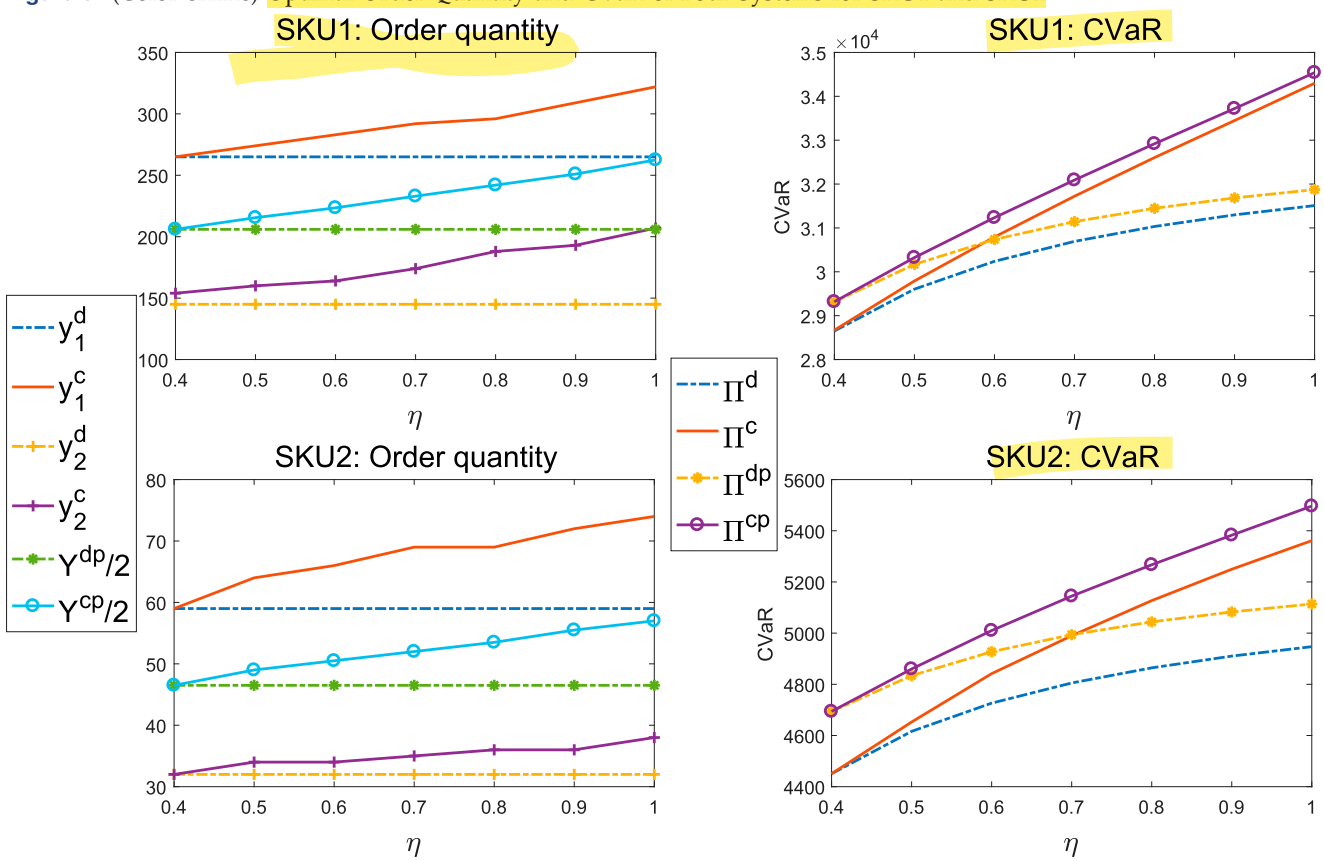
From Figure 5, we can see that for each SKU the optimal order quantities of system C are always larger than those of system D for both DCs, and the CVaR

Figure 4. (Color online) Biweekly Demand Data Points



of system C is also higher than that of system D; meanwhile centralization brings positive benefits for both SKUs when $\eta > 0.4 = \hat{\eta}$. These observations confirm part (a) of Proposition 1. Moreover, for SKU2, when $\eta = \hat{\eta} = 0.4$, the order quantities and the CVaR of system C are equal to those of system C, that is, $y_i^{\mathrm{d}} = y_i^{\mathrm{c}}$ for $i = 1, 2$, and $\Pi^{\mathrm{d}} = \Pi^{\mathrm{c}}$. Note that the profit margin of SKU2, $r = 0.5588$, and $\eta + (1-\eta r)^2 = 1.003 > 1$. This validates part (b) of Proposition 1. In addition, as the degree of risk aversion of the retailer decreases (η increases), the order quantities of the centralized systems increase whereas those of the decentralized systems are not affected, and the CVaR of all four systems increase. In particular, for each SKU,

Figure 5. (Color online) Optimal Order Quantity and CVaR of Four Systems for SKU1 and SKU2



there is a threshold such that when η is larger than this threshold, the CVaR of system C is higher than that of system DP, while when η is less than this threshold, the CVaR of system C is lower than that of system DP. Intuitively, when the degree of risk aversion of the retailer is close to that of the DC managers (η is small), the benefit of centralization (without pooling) is small; however, as the degree of risk aversion of the retailer decreases, the benefit of centralization significantly grows; and when the degree of risk aversion of the retailer is sufficiently less than that of the DC managers, that is, η is large enough, the benefit of centralization becomes larger than the benefit of inventory pooling. This is in line with the result of Theorem 1 (where the threshold is on $\hat{\eta}_i, i \in \mathcal{N}$ rather than η).

5.2. Comparative Statics

We further conduct extensive numerical studies using simulated parameters to complement our theoretical results by investigating the effects of profit margin, number of stores, and variance, correlation, and tail of demands on the optimal inventory levels and the corresponding CVaR values, as well as the strategic interplay between centralization and pooling (for system DP, again we focus on the results under the proportional allocation). Here, we only present a summary of the major observations, while providing the detailed results in Appendix B in the electronic companion.

- When the profit margin increases, the order quantities and the CVaR of all four systems increase. Moreover, as the profit margin increases, both the benefit of centralization and the benefit of inventory pooling increase.
- As the number of stores increases, the order quantity of system CP slightly decreases, whereas the order quantities of system C and system DP slightly increase. Meanwhile, as the number of stores increases, the CVaR linearly increases while the benefit of centralization and the benefit of pooling both increase.
- When the demand variance increases, the CVaR decreases, whereas the benefits of centralization and inventory pooling increase.
- The benefits of pooling for both decentralized and centralized systems decline as the correlation coefficient of demands increases while the CVaR of system D and system C are not affected much by the demand correlation.
- As the tail of the demand distribution becomes heavier, the benefit of pooling decreases whereas the benefit of centralization is not greatly affected.
- Centralization and inventory pooling are strategic substitutes; that is, the employment of one strategy offsets the value of the other: $\Pi^{cp} - \Pi^{c} \leq \Pi^{dp} - \Pi^{d}$.

6. Conclusion and Discussions

We study a multilocation risk-averse newsvendor model, where the stores are owned by a retailer and each location/store is operated by a manager who replenishes its stock to satisfy its own random demand. Both the retailer and the store managers are risk averse, and we adopt CVaR as the performance measure. We consider two alternative systems: a centralized system without inventory pooling and a decentralized system with inventory pooling. Under the framework of risk-averse decision-making, these two systems bring different benefits to the retailer. We find that when there is no inventory pooling, each store has a higher inventory level in the centralized system than in the decentralized system, and centralization brings positive benefits to the retailer as long as some store manager(s) is strictly more risk averse than the retailer. The magnitude of such benefit tends to increase as the risk preferences between the retailer and store managers become more misaligned. When there is inventory pooling, the inventory level in the decentralized system depends on how the residual profit is allocated to the store managers and is the outcome of the resultant ordering game among the managers. We characterize the Nash equilibrium of inventory level under the weighted proportional allocation rule. We find that as long as the store managers are sufficiently more risk averse than the retailer or the demands are very heavy tailed, the retailer will still prioritize centralization over inventory pooling. Furthermore, under some special cases, we derive bounds of the values of these two strategies and discuss some insights.

To complement our theoretical results, we use data from an online retailer in China to numerically illustrate and validate the main analytical findings of the paper. Moreover, by using simulated data, our numerical study further shows that the benefits of centralization and inventory pooling increase as the profit margin, the number of stores, or the demand variance increases; for positively correlated demands, it is more likely that centralization brings more benefit than inventory pooling; and centralization and inventory pooling are strategic substitutes. These results will provide some useful insights and guidelines for a retailer when considering to implement either centralization or inventory pooling. In our paper, due to the limited information on the decision-makers (DC managers), we have assumed certain values of their risk preference parameters $\hat{\eta}_i$ in our numerical study. Indeed, if we could obtain the information on the decision-makers' real-time demand (forecasting) models, inventory states and order quantities, we might be able to approximately estimate their risk preference parameters by running a regression model

based on the newsvendor fractile formula; that is, to compute the risk preference parameters that best fit the DC managers' ordering behavior. However, the issue of calibrating the risk preference parameters is beyond the scope of the current paper. Readers are referred to Kalinchenko et al. (2012) for how to calibrate such parameters using the regression model from option price data.

We have used CVaR in this paper as a measure of all agents' risk preferences due to its analytical tractability and the fact that agents' risk preferences can be completely ordered by a single parameter. Our main insight—centralization would generate more benefit than inventory pooling given there is sufficient discrepancy in the risk preferences of the retailer and the store managers—remains valid under the proportional allocation rule when we use exponential utility or entropic risk measures (the detailed analysis is given in Appendix C in the electronic companion). Nevertheless, unlike CVaR, the optimal order quantity under exponential utility is sensitive to scaling; that is, the optimal order quantity will be different if the profit is measured in cents as opposed to in dollars (see the appendix in Choi et al. 2011). Our choice of CVaR risk measure would be reasonable if the ordering decisions of the store managers are observed to be invariant to the unit of measurement (which could be easily tested experimentally). Similarly, our assumption that the retailer is less risk averse than the store managers, a commonly imposed assumption in principal-agent models, is not critical for the above insight. In Appendix C in the electronic companion, we show that under the proportional allocation, the benefit of centralization would still dominate if the retailer is much more risk averse than the store managers. That said, some of our results on system inventory levels do depend on the CVaR risk measure and the assumption that the retailer is less risk averse. For example, if the retailer is more risk averse, then centralization may no longer increase the inventory level in the system as postulated in Proposition 1, part (a). A systematic study of the optimal ordering decisions under general coherent risk measures would then be an interesting future research direction.

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Endnotes

[1] If one considers a more general store-dependent weight w_i, $i = 1, \ldots, n$, then the resultant allocation may no longer always be feasible.

[2] Note that the parameter ϱ in the Gaussian copula is not necessarily the coefficient of correlation of (X_1, X_2) unless the marginal distributions are also normal.

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